

Eye and Face Protection SOP

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1. Purpose

This SOP establishes requirements for eye and face protection to prevent injuries from flying particles, dust, molten metal splash, chemical splashes, harmful radiation (including welding arc flash and UV/IR exposure), and high-pressure fluid jets. Eye injuries are often irreversible, making strict adherence to this procedure critical wherever such hazards exist.

2. Scope

This procedure covers all operations involving grinding, cutting, welding, machining, chemical handling, compressed air or fluid use, and any task generating airborne particles or splash hazards. It applies to employees, contractors, and visitors within designated eye-protection mandatory zones marked with appropriate signage.

3. Types of Eye and Face Protection

- Safety spectacles with side shields, certified to IS 5983 / ANSI Z87.1, for general dust and particle protection in machine shops and assembly areas.
- Chemical splash goggles with indirect ventilation for handling acids, solvents, and other hazardous chemicals in the chemical storage and mixing areas.
- Welding helmets with auto-darkening filters matched to the correct shade number for the specific welding process (e.g., shade 10-13 for arc welding).
- Face shields worn over safety spectacles for grinding, chipping, and operations with a risk of larger flying debris.
- Laser safety eyewear rated to the specific wavelength for any laser-cutting or laser-marking operations.

4. Selection Based on Task

Task-specific risk assessments determine the required eye protection class before work begins. Supervisors must confirm that the correct protection type — not merely 'any eyewear' — is worn for each specific task, since spectacles alone do not protect against chemical splash, and goggles alone may not stop high-velocity fragments in grinding operations. Combination protection (goggles plus face shield) is mandatory for tasks with multiple hazard types.

5. Fit and Usage Guidelines

- Ensure the eyewear fits snugly against the face without gaps at the temples or brow.
- Employees who require corrective lenses must be issued prescription safety eyewear or over-the-glasses (OTG) safety goggles; personal prescription glasses alone are not compliant.
- Anti-fog coated lenses are provided for humid or temperature-variable environments to maintain visibility.
- Contact lens wearers in chemical-handling areas must use tightly sealed chemical goggles, as chemical vapors can become trapped beneath contact lenses.

6. Inspection and Cleaning

Eyewear must be inspected before each use for scratches, cracks, or loss of the protective coating that could impair vision or reduce protection. Lenses should be cleaned only with manufacturer-approved cleaning solutions and a microfiber cloth; abrasive materials that scratch the lens surface must not be used. Damaged eyewear must be reported to the Stores department and replaced before the next shift.

7. Emergency Response for Eye Exposure

In case of chemical splash or foreign particle contact with the eye, the affected employee must proceed immediately to the nearest eyewash station and flush the eye continuously for a minimum of 15 minutes while holding the eyelid open. The Site Medical Officer must be notified immediately, and the incident logged regardless of apparent severity, as some chemical burns manifest delayed symptoms.

8. Training and Compliance

All employees working in eye-hazard zones undergo mandatory training on hazard recognition, correct selection and fitting of protective eyewear, and eyewash station location and use. Non-compliance results in immediate work stoppage and is logged as a safety violation. Training is refreshed annually and after any eye-related incident.